

Due: Tuesday February 23rd, during class starting at noon

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Frames S and S' are coincident at time $t = t' = 0$. Frame S' moves with velocity v with respect to frame S , along the x (or x') direction. We have learned that measurements of events from the two frames, (x, y, z, t) and (x', y', z', t') , are related by the Lorentz transformation (LT)

$$x' = \gamma(x - vt) ; \quad y' = y ; \quad z' = z ; \quad t' = \gamma(t - vx/c^2)$$

1. The constancy of the speed of light demands that

$$x'^2 + y'^2 + z'^2 - c^2 t'^2 = x^2 + y^2 + z^2 - c^2 t^2$$

This will be true irrespective of the direction of the relative velocity between the frames. For our particular case of relative velocity in the x (x') direction, this reduces to

$$x'^2 - c^2 t'^2 = x^2 - c^2 t^2$$

Prove this equation algebraically, using the LT.

2. Let's omit the perpendicular directions (y, y', z, z') for the rest of this problem set.
 - (a) Write down the inverse LT, i.e., (x, t) in terms of (x', t') , using the fact that frame S moves with velocity $-v$ with respect to frame S' .
 - (b) Invert the LT algebraically to find the inverse LT, i.e., determine (x, t) in terms of (x', t') algebraically.
3. Using the rapidity $\phi = \tanh^{-1}(v/c)$, the LT can be written as

$$\begin{pmatrix} x' \\ ct' \end{pmatrix} = \begin{pmatrix} \cosh \phi & -\sinh \phi \\ -\sinh \phi & \cosh \phi \end{pmatrix} \begin{pmatrix} x \\ ct \end{pmatrix} = \Lambda(\phi) \begin{pmatrix} x \\ ct \end{pmatrix}$$

Since hyperbolic functions are important for the LT, look up the standard expressions for $\sinh(\phi_1 + \phi_2)$ and $\cosh(\phi_1 + \phi_2)$.

- (a) Show that two successive LT's, with rapidities ϕ_1 and ϕ_2 , result in a LT of rapidity $\phi_1 + \phi_2$. In other words, show via matrix multiplication that

$$\Lambda(\phi_1)\Lambda(\phi_2) = \Lambda(\phi_1 + \phi_2)$$

- (b) An LT at rapidity $-\phi$ should be the inverse of the LT at rapidity ϕ . Thus we should have the matrix relationship

$$\Lambda(-\phi) = [\Lambda(\phi)]^{-1} \quad \text{or} \quad \Lambda(-\phi)\Lambda(\phi) = I$$

where I is the 2×2 identity matrix. Show by matrix multiplication that $\Lambda(-\phi)\Lambda(\phi)$ is indeed the identity matrix. You might find it helpful to remember that $\cosh^2 u - \sinh^2 u = 1$.